Association for Clinical Genomic Science (ACGS) guidelines for the classification of oncogenicity of somatic variants in cancer - recommendations by the UK somatic variant interpretation group (SVIG-UK)

SUPPLEMENTARY MATERIAL

Table of Contents

Supplementary Table 1 – SVIG-UK Evidence Criteria for Oncogenic Status of Somatic Variants

Evidence criteria		Evidence strength
O1	**SVIG-UK Canonical Variants List (Supplementary Table 3)**	Stand-alone oncogenic
O2 / O2 (RNA)	**Null variant in a tumour suppressor gene (TSG)**	Very strong [+8] Strong [+4] Moderate [+2] Supporting [+1]
	Guidance: This code may be applied for null variants (nonsense, frameshift, canonical ±1 or 2 splice sites, initiation codon, single or multi-exon deletion) in a gene known to act as a TSG in the cancer type under investigation. **O2 is applied according to the PVS1 decision tree in Abou Tayoun *et al*., (2018)[1] and in line with Horak *et al*., (2022)[2].** Please note, Abou Tayoun *et al*., (2018) decision tree refers to PVS1, which is an equivalent to O2 (i.e. PVS1 > PVS1_Strong > PVS1_Moderate > PVS1_Supp is equivalent to O2_VSTR > O2_STR > O2_Mod > O2_Supp respectively). **O2 is specific to variants where LOF is the known disease mechanism**. If a gene can predispose to disease via both loss-of-function (LOF) and gain-of-function (GOF) mechanisms and an O2-eligible variant leads to a predicted (or experimentally observed) aberration expected to result in GOF, O9 should be used as opposed to O2[3]. Where available, also refer to gene-specific guidance, including *BRCA1*/*BRCA2* and MMR genes (CanVar/CanVIG-UK) and *RUNX1* (ClinGen VCEP). <u>**Start-loss/initiation codon variants**</u> For variants impacting the initiation codon and resulting in start-loss, start re-initiation may occur at an alternate ATG or non-ATG start-site either downstream or even upstream of the original initiation codon. Where alternative functional gene transcripts (i.e. those found in transcript or expression-based databases) using a different initiation codon exist, then O2 should NOT be applied at any level of strength.[1] If no alternative functional transcripts utilising a different initiation codon exist, search for the next putative in-frame initiation codon (methionine) downstream. O2 can be applied at a moderate strength if one or more (likely) oncogenic variants exist 5' of the putative downstream initiation codon. If no (likely) oncogenic variants exist 5' of the putative downstream initiation codon, then O2 should be applied at supporting (see Supplementary Figure 3).[1] <u>**Stop-gain variants within the first 100 bp of the first exon**</u> Variants that create a STOP in the first 100 bp of the first exon are likely to evade nonsense mediated decay (NMD) as re-initiation of translation may occur 3' of the STOP using an alternative start codon resulting in a functional (or semi-functional) protein[4] . If an alternative initiation codon is identified, consider whether to apply O2 at an appropriate strength by assessing the lost N-terminal region according to the principles described in the Abou Tayoun *et al*.,	

(2018) decision tree[1] (i.e. is the missing region critical to normal function/does it account for >10% of the entire protein?).

If no alternative in-frame start codon is identified apply O2 at maximum strength (O2_VSTR [+8]).

Nonsense/frameshift variants (generating a STOP after the first 100 bps)

These will always follow the Abou Tayoun *et al*., (2018) nonsense or frameshift section of the PVS1 decision tree[1].

The position of the nonsense variant or resulting termination codon caused by a frameshift variant (extreme 3' end of the TSG gene under review) may result in NMD escape or an overall intact protein with reduced deleterious effect.

The position of the STOP should be noted; if the premature termination codon (PTC) occurs in the last 50 bp's of the penultimate exon or in the final exon, then NMD would generally not be expected (see relevant gene-specific guidelines for exceptions to this rule).

Stop-loss variants

For frameshift/stop-loss variants, prediction of non-stop mediated decay (NSD) is dependent on the presence of a novel, in-frame termination codon in the 3' UTR. If a novel, in-frame termination codon does not exist between the abolished wild type canonical termination codon and the polyA site sequence (AATAAA) in the 3' UTR, the ribosome runs into the polyA sequence and will not dissociate and this will trigger NSD.

- **In the absence of a novel, in-frame termination codon** in the 3' UTR, the mRNA transcript is likely to undergo NSD; apply O2_VSTR [+8].
- **If there is a novel, in-frame termination codon** within the 3'UTR upstream of the polyA signal, the predicted consequence is a protein with additional amino acids (i.e. normal protein sequence but extended and NSD is not predicted); apply O9_mod [+2] (see Supplementary Figure 3).

For frameshift variants with extension of sequence and an alteration to the normal wild-type coding sequence where neither NSD or NMD are expected, apply O2 at the relevant strength level according to Abou Tayoun *et al*., (2018) (i.e. O2_STR or O2_mod depending on proportion of protein affected and functional significance of altered region).

Canonical splice variants (splice donor/acceptor +/- 1, 2 dinucleotide variant)

When assessing canonical splice site variants, unless there are RNA studies to show the effect on the protein, it is assumed that the exon it is closest to is skipped which may result in an in-frame or out-of-frame effect; determine the effect and apply O2 at a relevant strength according to Abou Tayoun *et al*., (2018)[1] and Figure 2 of Walker *et al*., (2023)[3].

When assessing the effect on the reading frame it is also important to look for nearby (suggest up to approximately +/- 20 bp[1]) cryptic splice sites or newly generated splice sites using splice predictive tools (e.g. SpliceAI). It is recommended that the lowest strength of O2 should be applied when different in-frame and out-of-frame scenarios are possible, however, once RNA studies have been performed this over-rules all possible predicted effects and O2 (RNA) should be applied accordingly (see section on Splicing Assays below).

The impact of canonical splicing variants affecting the first and final intron splice donor/acceptor sites can be difficult to predict; see Figure 2 of Walker *et al.*, (2023) for O2 (PVS1) weighting recommendations.

For canonical splice variants predicted to result in an in-frame transcript or to use an alternative cryptic in-frame splice site, additional evidence is required to ascertain if the region lost is critical to normal protein function (see additional notes section).

+2T>C splice variants may result in functional GC splice sites, therefore, O2 should be used cautiously in the absence of RNA studies[5]. Use of SpliceAI is recommended to assess the likely impact on splicing; PVS1 should not be applied for +2T>C variants with SpliceAI delta score <0.8 in the absence of RNA evidence. In addition, approximately 1.5% of natural splice sites use a C at +2 (GC splice site) therefore a +2C>T variant is likely to increase the efficiency of the splice site and O2 is not applicable - use SpliceAI to confirm prediction.

Splicing Assays [O2 (RNA)]

Recently the Clinical Genome Resource (ClinGen) Sequence Variant Interpretation (SVI) Splicing Subgroup[3] published recommendations which included interpretation of experimental data demonstrating an abnormal splicing profile using PVS1_*strength* (RNA) and the PVS1 Abou Tayoun *et al*., (2018) decision tree. Accordingly, **the O2 (RNA) code should be used in place of O10 where somatic variants are shown to impact splicing by RNA- or cDNA-based assays, even when O2 would not ordinarily be applied based on variant location or type.** The experimental data must first be assessed to determine whether the assay performed was of sufficient quality to be used as evidence supportive of the application of the O2 (RNA) code. Considerations include assay design, RNA source, methodology, technology, quantification measurement and use of controls; a more comprehensive list of considerations is included in Table S9, Walker *et al.,* (2023). Additional validation requirements are available in the CanVIG-UK Consensus Specification for Cancer Susceptibility Genes (v3.0) [6].

Where there is sufficient confidence in the splicing assay for application of O2 (RNA), the weighting at which this code can be applied can be determined based on the observed impact on splicing according to Figure 1 in Abou Tayoun et al., (2018)[1] and the level of incomplete ("leaky") splicing[3]. Defining thresholds for incomplete splicing is complex and requires gene/exon-specific disease knowledge. In the absence of gene/exon-specific data, refer to the CanVIG-UK Consensus Specifications[6] for relevant/acceptable thresholds of 'leakiness'.

The recommendations provided by Walker *et al*. (2023) advise upgrading the O2 (RNA) equivalent code (PVS1) to *very strong* for in-frame RNA skipping variants encompassing undisputed clinically relevant residues[3].

O2 (RNA) should not be used for non-canonical splice variants in the absence of functional data.

Additional notes

For variants that require evidence of "**region critical to protein function**", the presence of loss-of-function missense variants that are classified as (likely) oncogenic (or equivalent using in-house guidelines prior to the release of SVIG-UK guidelines) within the affected region can be used to upgrade from moderate to strong (assuming they are not acting on splicing). Consideration of protein domains, sequence conservation and constraint may also be informative, however, please note these should not be used in isolation as evidence for "region critical to protein function".

	Use caution if a splice variant leads to expression of a well-known alternative isoform that preserves tumour suppressor functionality or if in the case of exon skipping (in-frame) where the protein is relatively intact. For variants not undergoing NMD, functional evidence may be used to uplift the weighting of O2 but O10 should not be used with O2_Very Strong [+8 points]. **Equivalent germline code (Richards *et al.*, 2015)[7]:** PVS1 / PVS1_RNA **Evidence criteria combination guidance is detailed in Supplementary Figure 1. Non-permitted combinations and restrictions exist for this criterion.**	
O3	**Absent or very rare in population database**	
	Zero observations of the variant across all populations in gnomAD v4.1	Moderate [+2]
	Variant is not absent but is at an allele frequency of ≤0.001% (≥100,000 alleles) in gnomAD V4.1	Supporting [+1]
	Guidance: O3 can be used where variants are absent across all populations from population databases; the Genome Aggregation Database (gnomAD) v4.1 is the preferred database at this time. O3 may also be applied in the situation that the variant is sufficiently rare within gnomAD. Scientific judgement should be applied; however, in the absence of constraint calculation for somatic variants, an allele frequency threshold of ≤0.001% (in any sub-population consisting of ≥100,000 alleles) in gnomAD v4.1 is recommended. Where a variant is present in >1 allele in any sub-population consisting of <100,000 alleles, O3 should NOT be applied. However, if the somatic variant is in a gene known to cause predisposition to hereditary cancer, ClinGen variant curation expert panel (VCEP) gene-specific guidelines must be consulted (if they exist) to establish a cut-off that takes disease prevalence into account. It is important to check that the variant position is covered to sufficient read depth in gnomAD. GnomAD coverage data is available from, https://gnomAD.broadinstitute.org/downloads#v4-coverage (v4.1). Apply caution when using O3 for complex and/or large insertion/deletion variants which may not be reliably identified by NGS technology or may have been called using alternative terminology. It may be pertinent to ascertain whether other indels have been detected within the region. Somatic mosaicism of variants in genes such as (but not limited to) *DNMT3A*, *ASXL1*, *TET2* and *TP53* during hematopoietic clonal expansion (clonal haematopoiesis of indeterminate potential or CHIP) can occur with ageing in healthy individuals[8,9]. In such genes, the 'allele balance' chart in gnomAD can be used to assess whether entries in the gnomAD dataset may represent CHIP contamination. If the variant under investigation does not meet any population rule codes (O3 or B1) AND has a total allele frequency >0.00001 (0.001%) then low variant allele frequency (VAF) alleles (which are likely to represent CHIP contamination in the database) can be excluded and the total allele frequency based on the number of alleles with VAF between 0.3 and 0.65 can be re-calculated. The re-calculated frequency can then be used to assess whether O3 is now applicable. Apply scientific judgement when assessing the relevance of CHIP contamination for the gene variant under investigation.	

	O3 (absence in controls) provides evidence of rarity, rather than evidence against benignity and therefore should not be used if all other evidence suggests that a variant is likely benign and application of O3 would move it into the VUS category. SVIG-UK acknowledges the recommendation of the ClinGen SVI Working Group (2020) to downgrade the weight of PM2 (O3) to "supporting" by default (SVI Recommendation for Absence/Rarity (PM2) - Version 1.0). However, after evaluating the impact of this recommendation within the SVIG-UK framework, as well as reviewing the ACGS mini-impact assessment on the effect of downgrading PM2 and considering the proposed changes to PP3/BP4[10], the SVIG-UK group has decided to retain the application strength of PM2 up to moderate. This decision is consistent with current UK guidance on germline variant interpretation[6]. **Equivalent germline code (Richards *et al.*, 2015)[7]:** PM2 **Evidence criteria combination guidance is detailed in Supplementary Figure 1. Non-permitted combinations and restrictions exist for this criterion.**
O4	**Enriched in a somatic variant database (compared to prevalence in controls*)**

Criterion	Strength
Missense and splice variants	
>10 'on-target' (and/or 'off-target' with confirmed consistent mechanism of action†) entries in an international (e.g. GENIE) or national (multicentre) curated database	Strong [+4]
5-10 'on-target' (and/or 'off-target' with confirmed consistent mechanism of action†) entries in an international (e.g. GENIE) or national (multicentre) curated database	Moderate [+2]
>10 VUS 'on-target' (and/or 'off-target' with confirmed consistent mechanism of action†) entries in a curated in-house database	
5-10 VUS 'on-target' (and/or 'off-target' with confirmed consistent mechanism of action†) entries in a curated in-house database	Supporting [+1]
Frameshift (truncating) and nonsense variants	
>50 frameshift (truncating) or nonsense 'on-target' (and/or 'off-target' with confirmed consistent mechanism of action†) entries in an international (e.g. GENIE) or national (multicentre) curated database at the same position or further downstream	Strong [+4]
20-50 frameshift (truncating) or nonsense 'on-target' (and/or 'off-target' with confirmed consistent mechanism of action†) entries in an international (e.g. GENIE) or national (multicentre) curated database at the same position or further downstream	Moderate [+2]
10-19 frameshift (truncating) or nonsense 'on-target' (and/or 'off-target' with confirmed consistent mechanism of action†) entries in an international (e.g. GENIE) or national (multicentre) curated database at the same position or further downstream	Supporting [+1]
In-frame deletions/insertions	
>50 in-frame deletions/insertion entries within the deleted region in an international (e.g. GENIE) or national (multicentre) curated database	Strong [+4]
20-50 in-frame deletions/insertion entries within the deleted region in an international (e.g. GENIE) or national (multicentre) curated database	Moderate [+2]
10-19 in-frame deletions/insertion entries within the deleted region in an international (e.g. GENIE) or national (multicentre) curated database	Supporting [+1]

Guidance:

*Application of O3 (absence/rarity in a population database) is a prerequisite for applying O4.

†This criterion has been designed to be **tumour agnostic**, however, to prevent counting of entries detected in tumours that do not match the mode of action of the tumour type under assessment, case-counting can be limited to 'on-target' tumour types. For example, *EZH2* may act as an oncogene in many solid tumours and lymphomas, while acting as a tumour suppressor in myelodysplastic syndromes, acute myeloid leukaemia (although rare) and T-cell leukaemias[11–13].

'On-target' may refer to a specific tumour type or to the wider clinical indication. For example, when assessing a variant detected in a patient with acute myeloid leukaemia (AML), 'on-target' entries may be specific to AML, to myeloid disorders or to all haematopoietic disorders (myeloid and lymphoid). The definition of an 'on-target' tumour type is currently at the discretion of individual laboratories and may differ depending upon the tumour type and gene being assessed. Entries in 'off-target' tumour types may be combined with 'on-target' tumour types where the mechanism of action in both the 'on-target' and 'off-target' tumour types is consistent.

International, national or local databases may be independently used to demonstrate enrichment, but local entries must not be combined with other datasets (i.e. GENIE/COSMIC). The thresholds set for criteria weighting from in-house databases reflects any potential bias introduced by data from single centre sources, technology and bioinformatics pipelines. Where the variant is present in multiple datasets (i.e. literature, international/national (multicentric) curated database and/or in-house database), the dataset giving the strongest level of evidence should be used. Evidence from different data sets should not be combined/stacked (e.g. where O4_moderate [2 pts] from GENIE dataset and O4_moderate [2 pts] from national (multicentre) database the maximum weighting that can be used for this code is O4_mod [2 pts]).

The GENIE database contains multiple samples from a single patient (i.e. multiple tumours from a single patient or repeat samples from a single patient at different time points) and therefore care should be taken to ensure that multiple samples from the same patient are only counted as a single entry. When searching the GENIE database, the total number of variants and number of duplicate variants can be found at the bottom of the page. It is recommended that the most recent 'GENIE Cohort' dataset available on cBioportal is used (https://genie.cbioportal.org/#summary).

Please note if COSMIC is used for enrichment analysis, much higher thresholds should be applied for solid tumours (e.g >50 solid tumour entries for missense variants to achieve a strong strength). This is to account for known differences in the COSMIC database including higher representation of solid tumours, sample duplication and contamination with constitutional variants.

Published peer-reviewed literature can also be used for case counting using the stated thresholds above. Please do not combine published cases with above databases to avoid the risk of double counting.

Case control data is currently unavailable for most large somatic databases (such as GENIE or COSMIC), and so recommendations for enrichment are based upon expert consensus rather than statistical modelling.

	Missense variant case counting Enrichment for missense variants, as per Froyen *et al.*, (2019), should be based on the number of entries for the same amino acid change as the variant under investigation, regardless of the underlying nucleotide change. For example, database entries for p.Val12Leu can be counted and combined regardless of whether this amino acid substitution was the result of c.34G>C or c.34G>T. **Splice variant case counting** For splice variants it is important to note that the impact on splicing is determined by the nucleotide sequence and therefore **enrichment for splicing variants must be based upon the exact nucleotide change**. Do not apply for different nucleotide change affecting the same position even if predicted to have similar impact. **Frameshift (truncating) and nonsense variants** Caution is required for case counting downstream frameshift/nonsense variants that exert their effect via an alternative mechanism of action, for example where a LOF or GOF mechanism is dependent upon the variant location within the gene. When case counting, exclusion of variants which do not match the disease mechanism of the variant under investigation should be considered. Caution should be applied when using O4 for variants that create a STOP in the first 100 bp of the first exon. These are likely to evade nonsense mediated decay (NMD), as re-initiation of translation may occur 3' of the STOP using an alternative start codon resulting in a functional (or semi-functional) protein[4]. **If no alternative in-frame start codon is identified;** apply O4 at appropriate strength as indicated above. **In-frame insertion-deletion variant case counting** For insertion variants, only those that exactly match the variant under assessment (VUA) should be counted i.e. variants resulting in insertion of alternative amino acids at the same position cannot be counted. For deletions, only those that exactly match the VUA or are smaller than and within the deleted region of the VUA should be counted.	
	Equivalent germline code (Richards *et al.*, 2015)[7]: PS4 **Evidence criteria combination guidance:** ● If B1 is applied (at any level) then O4 should not be used. ● O4 cannot be applied if B2 is applied	
O5	**Variants that affect same location and /or result in a similar impact**	
	Splicing variants (see O5 guidance section for weightings and prerequisites) **Initiation codon variant** where a different nucleotide substitution affecting the initiation codon has been determined to be oncogenic	Strong [+4]

	Different amino acid change (higher REVEL score): Novel missense change at an amino acid residue where a different amino acid change determined to be (likely) oncogenic has been seen before <u>and</u> the variant under examination has an <u>equivalent</u> (difference between the scores is ≤0.02) or <u>more</u> deleterious REVEL score than the reference variant <u>or</u> both reference variant and variant under examination have REVEL scores ≥0.773[10]. **Splicing variants (see O5 guidance section for weightings and prerequisites)** **Initiation codon variant** where a different nucleotide substitution affecting the initiation codon has been determined to be <u>likely</u> oncogenic.	Moderate [+2]
	Different amino acid change (lower REVEL score): Novel missense change at an amino acid residue where a different amino acid change determined to be (likely) oncogenic has been seen before where the variant under examination has a REVEL score of >0.7 and <0.773[10] <u>and is less</u> deleterious than the reference variant. **Splicing variants (see guidance section below for weightings and prerequisites) (see table below for weighting).**	Supporting [+1]

Guidance:

Missense variants considerations

Reference variant must have been classified as (likely) oncogenic according to SVIG-UK guidelines (or equivalent using in-house guidelines prior to the release of SVIG-UK guidelines) by an accredited laboratory.

Mechanism of action (i.e. LOF or GOF) for the variant under investigation must be the same as the reference variant.

Splicing variant considerations

O5 can be used for splicing variants where a different nucleotide substitution has been classified as (likely) oncogenic and the variant being assessed is predicted by *in silico* tools (i.e. SpliceAI) to have a similar or greater deleterious impact on the mRNA. Reference variants must have been classified as (likely) oncogenic according to SVIG_UK guidance (or equivalent using in-house guidelines prior to the release of SVIG-UK guidelines) by an accredited laboratory.

The following table is adapted from Walker et al., 2023[3]. Please refer to the paper for further guidance (note PS1 should be considered equivalent to O5 in this context).

Variant under assessment (VUA)	**Computational/predictive code for VUA**	**Position of comparison variant relative to VUA**	**O5 code applicable to VUA**	
			With Oncogenic comparison variant	**With Likely Oncogenic comparison variant**
Located outside splice donor/acceptor (+/-1,2 dinucleotide position)	O6	Same nucleotide	O5_str	O5_mod
	O6	Within same splice donor/acceptor motif (incl. +/- 1,2 positions)	O5_mod	O5_supp
Located at splice donor/acceptor (+/-1,2 dinucleotide position)	O2_vstr	Within same splice donor/acceptor +/-1,2 dinucleotide positions	O5_supp	N/A
	O2_vstr	Within same splice donor/acceptor motif but outside +/-1,2 dinucleotide	O5_supp	O5_supp

	O2_str, O2_mod or O2_supp	Within same splice donor/acceptor +/-1,2 dinucleotide	O5_str	N/A
	O2_str, O2_mod or O2_supp	Within same splice donor/acceptor motif but outside +/-1,2 dinucleotide	O5_mod	O5_supp

Splice region donor motif is defined as last 3 bases of the exon and 3-6 nucleotides of intronic sequence adjacent to the exon. Splice acceptor motif is defined as first base of the exon and 1-20 nucleotides upstream from the exon boundary[3].

Application of O5 in the context of splicing predictive tools [either O6 or O2 (RNA)] should demonstrate a predicted impact on splicing [i.e. O2 (RNA) can be applied in combination with O5 or, alternatively, O6 can be applied in combination with O5].

Prerequisite for all splice variants: the predicted event of the VUA must precisely match the predicted event of the comparison (likely) oncogenic variant (i.e. both lead to exon skipping or both lead to enhanced use of a cryptic splice motif) and both the reference variant and the variant under investigation should have SpliceAI scores >0.2. For exonic splice region variants, predicted or functional effect of the missense variant should be considered before applying this code.

Equivalent germline code (Richards *et al.*, 2015)[7]: PM5

Evidence criteria combination guidance is detailed in Supplementary Figure 1. Non-permitted combinations and restrictions exist for this criterion.

O6	**Computational evidence supports a deleterious effect on the gene or gene product (conservation, evolutionary, splicing impact etc).**	Supporting [+1]

Guidance:

Missense variants:

- For predicting the impact of missense variants, it is recommended that a meta-predictor tool, such as REVEL, is used[14].
- A REVEL score of ≥0.7 provides evidence of oncogenicity and O6 can be applied at a SUPPORTING level[†15]. For GOF genes, O6 can still be applied but should be limited to variants within functional regions where missense variants are known to confer GOF.

[†]Threshold based on analysis of a cohort of 321 pathogenic (ACMG 4/5) and 1238 benign (ACMG 1/2) variants classified by the Leeds Genetics Laboratory to achieve a sensitivity of ~95%. This threshold is also supported by the study of Cubuk *et al.*, (2021)[16] and is consistent with the findings of Pejaver *et al.*, (2022)[10].

Splicing variants:

- *In silico* splicing prediction tools can be used as evidence to support a significant impact on splicing potential for splice site variants outside the canonical splice acceptor (-1 and -2) and donor (+1 and +2) regions.
- O6 may also be applied where splice prediction algorithms indicate the introduction of a cryptic splice site with the potential to cause aberrant splicing, e.g. the introduction of a 3’ (acceptor) site in an intron.

	• The use of SpliceAI is recommended and O6 may be used at a SUPPORTING level for variants with a donor/acceptor loss or gain Δ score ≥0.2[3,17,18]. Where gene-specific VCEP guidance is available, thresholds from these guidelines can be used; however, O6 can only be applied at a maximum strength of SUPPORTING regardless of gene-specific guidance permitting the use of the equivalent code at higher levels of strength. **Equivalent germline code (Richards *et al.*, 2015)[7]:** PP3 **Evidence criteria combination guidance is detailed in Supplementary Figure 1. Non-permitted combinations and restrictions exist for this criterion.**	
O7	**Located in a mutational hotspot and/or critical and well-established functional domain (e.g. active site of an enzyme) without benign variation**	
	Variant under assessment (VUA) at a mutational hotspot in cancerhotspots.org with ≥50 entries at the same amino acid position, of which there are **at least 10 entries for the same amino acid change.**	Strong [+4]
	VUA at a mutational hotspot in cancerhotspots.org with <50 entries at the same amino acid position, of which there are **at least 10 entries for the same amino acid change** OR Meets criteria for moderate application according to guidance below	Moderate [+2]
	2-9 entries for the same amino acid change as the VUA in cancerhotspots.org database [19] OR Meets criteria for supporting application according to guidance below	Supporting [+1]
	Guidance: This code can be applied for **very specific residues described in the Cancer Hotspots database** (http://www.cancerhotspots.org/#/home[20,21]), where the variant is **consistent with the mode of action** of the gene for the cancer type under investigation (i.e. B1 has NOT been applied). O7 may also be applied at **moderate level** for missense and small in-frame del/ins variants† where **all of the following criteria are applicable***: 1. Absence of local benign variation (i.e. absence of variants in gnomAD) 2. Evidence for local enrichment of oncogenic missense or in-frame ins/del variants 3. Variant located in a well-established functional domain (e.g. active site of an enzyme) †The definition of a 'small in-frame deletion/insertion' is at the discretion of the analyst as this is variable depending upon the gene and location and should be consistent with other variants reported in the literature. *Where **two of the three criteria** above are applicable **but there is either** (1) limited local benign variation, (2) limited evidence for local enrichment of oncogenic variants, or (3) the variant is in a less well characterised region/domain; **O7 should not be used above supporting level**.	

<table>
<tr><td></td><td colspan="2">Alternatively, O7 may be applied at a moderate level if the variant impacts a well-established critical residue based on protein-to-protein paralogs of oncogenic variants (e.g. critical residue within a highly conserved essential consensus sequence, example; one of the 8 key Cys or His residues in the C3HC4 RING finger domain consensus sequence in the CBL gene) providing O3 is also applicable (at either a moderate or supporting level).

O7 may be applied at a supporting level where there is an absence of benign variation and where in silico protein structural models predicts a deleterious structural or functional impact in a region of the protein with well-established function.

The literature, Alamut and online resources such as Decipher (https://www.deciphergenomics.org/) may assist with the identification of functional domains and hotspots. Useful variant plots displaying disease-associated variants within a region are available via the GENIE and ClinVar databases. VCEP gene-specific guidelines may also be referred to for known mutational hotspots/critical domains but may require gene-specific knowledge for the applicability of use in the context of an acquired variant.

Since functional studies frequently inform delineation of critical domains/hotspots, caution should be applied when using O7 together with O10 to avoid any overlap. It is therefore recommended that O7 can only be used alongside O10 if the hotspot/critical region can be defined using evidence independent of functional studies

The use of alternative relevant evidence not already considered in these guidelines is permissible for the application of O7; however, it is essential that scientific judgement is applied to determine the strength of application and to prevent double counting evidence across multiple codes.
<hr>Equivalent germline code (Richards et al., 2015)[7]: PM1/PP2

Evidence criteria combination guidance is detailed in Supplementary Figure 1. Non-permitted combinations and restrictions exist for this criterion.</td></tr>
<tr><td>O8</td><td>Overall constraint for missense variation, at the level of the gene or domain/region, where missense variants are a common mechanism of disease.</td><td>Supporting [+1]</td></tr>
<tr><td></td><td colspan="2">Guidance:

GnomAD missense constraint scores should be used to demonstrate missense constraint at the level of the gene (gnomAD Z ≥3.09). Where data exists defining regional constraint scores this should be used in place of gene level data; as such O8 should not be used if there is evidence that the variant is within a region demonstrating tolerance to missense variation. Regional missense constraint scores are available via DECIPHER or gnomAD (v2.2.1) missense constraint tracks. The MetaDome web server (https://stuart.radboudumc.nl/metadome/) also provides a regional Tolerance Landscape visualisation for proteins.

Where data exists defining regional enrichment for oncogenic variants, this should be used in place of evidence of missense constraint (i.e. O7 in place of O8).
<hr>Equivalent germline code (Richards et al., 2015)[7]: PP2

Evidence criteria combination guidance is detailed in Supplementary Figure 1. Non-permitted combinations exist for this criterion.</td></tr>
<tr><td>O9</td><td colspan="2">Protein length change due to in-frame deletions/insertions in a non-repeat region and stop-loss variants or truncating variant in the final exon of an oncogene predicted to result in a gain-of-function</td></tr>
</table>

<table>
<tr><td rowspan="3"></td><td>In-frame del/ins of >1 amino acid (< 1 exon) within a known functional domain

or

Stop-loss variant resulting in extension of the protein and use of an in-frame stop codon in the 3’ UTR where non-stop decay (NSD) is <u>not</u> predicted (see Supplementary Figure 3)

or

Truncating variant in the final exon of an oncogene predicted to result in a gain-of-function[3].</td><td>Moderate [+2]</td></tr>
<tr><td>In-frame del/ins of single amino acid affecting residue within a known functional domain</td><td>Supporting [+1]</td></tr>
<tr><td colspan="2">Guidance:

Where available, application should be in accordance with gene-specific ClinGen VCEP guidelines.

O9 should not be used if a splicing assay [O2 (RNA) applied] has been used to demonstrate protein length change, (i.e. in the case of small in-frame insertion/deletion events <1 exon and stop-loss variants resulting in increase in protein length).
<hr>Equivalent germline code (Richards et al., 2015)[7]: PM4

Evidence criteria combination guidance is detailed in Supplementary Figure 1. Non-permitted combinations exist for this criterion.</td></tr>
<tr><td rowspan="4">O10</td><td colspan="2">Well-established in vitro or in vivo functional studies demonstrating a functionally abnormal result consistent with the mechanism of disease</td></tr>
<tr><td colspan="2">Protein function</td></tr>
<tr><td>In accordance with the CanVIG-UK Consensus Specification[6] the weighting of O10 should be determined according to the Clinical Genome Resource (ClinGen) Sequence Variant Interpretation (SVI) recommendations[22].</td><td>Very strong [+8]
Strong [+4]
Moderate [+2]
Supporting [+1]</td></tr>
<tr><td colspan="2">Guidance:

This code may be applied where in vivo or in vitro functional studies support an oncogenic consequence of the specific variant in the cancer type under investigation (e.g. supportive of an activating effect in an oncogene or of a null effect in a TSG).

This code is applied at variant level only. Where functional data, for example from biochemical testing, provides support at the gene rather than variant level, this cannot be used in the variant biological classification (but may be incorporated when assessing clinical actionability).

In the absence of gene-specific guidance, O10 should be applied according to the CanVIG-UK Consensus Specification[6] where the germline evidence code PS3 should be considered analogous to the SVIG-UK evidence code O10. In accordance with this, clinical validity of functional evidence and consequential weighting of O10 should be determined according to ClinGen SVI recommendations[22].

Specific points of note taken from the CanVIG-UK Consensus Specification[6]:</td></tr>
</table>

	Protein function: ● Variant is considered to have functionally abnormal effect if protein activity assay or functional impact is <25% of wildtype level. ● If the functional assay demonstrates an intermediate level of impact (i.e. the functional impact on the protein does not reach the threshold for a functionally abnormal effect), O10 evidence can be downgraded by one or more levels. This decision should be guided by consideration of how closely the assay reflects the known disease mechanism and/or in accordance with gene-specific guidance. ● **In the case of availability of data from multiple assays:** when considering assays of variable evidence strength and/or combining concordant/discordant functional assay results, use the guidance and combination tables described within the CanVIG-UK Consensus Specification[6]. ● Evidence should refer to biologically relevant transcripts. **Use of RNA splicing assays are incorporated into O2 (RNA).**	
	Equivalent germline code (Richards *et al.*, 2015)[7]: PS3 **Evidence criteria combination guidance is detailed in Supplementary Figure 1. Restrictions exist for this criterion.**	
O11	**Tumour phenotype**	Moderate [+2] Supporting [+1]
	Guidance: Application should be in accordance with gene-specific ClinGen VCEP guidelines (where available) but limited to the level of strength as permitted in these guidelines. It is recommended that 'stacking' of evidence within this criterion should be avoided, for example, where there are multiple cases of an exonuclease domain *POLE* variant with high tumour mutation burden (TMB) the evidence should be considered concordant and O11 criterion be applied at a maximum strength of O11 supporting. This criterion enables findings related to the tumour phenotype, including those from non-genetic disciplines, to be used as evidence in the classification of somatic variants. Supplementary Table 4 provides further details and examples. Supplementary Table 4 is not an exhaustive list. As this is a new line of evidence not previously utilised in somatic variant interpretation, scientific judgement must be applied, and a more cautious approach is currently advised.	
	Equivalent germline code (Richards *et al.*, 2015)[7]: None	

Supplementary Table 2 – SVIG-UK Evidence Criteria for Benign Status of Somatic Variants

<table>
<tr><th colspan="2">Evidence Criteria</th><th>Evidence Strength</th></tr>
<tr><td rowspan="4">B1</td><td colspan="2">Variant is present at a high level in a general population database</td></tr>
<tr><td>Variant is present at >1% (or as specified by respective expert panel group)</td><td>Stand-alone benign</td></tr>
<tr><td>Variant is present at >0.1%</td><td>Strong [-4]</td></tr>
<tr><td colspan="2">Guidance:

Note the overall allele frequency and the highest population allele frequency listed in gnomAD; if either the overall or the highest from any five general continental populations; African/African American, East Asian, European (non-Finnish), Admixed American and South Asian is >0.1% (in >1000 individuals) then apply B1 at a level of strength according to the table above[23].

If the somatic variant is in a gene known to cause predisposition to hereditary cancer, ClinGen VCEP gene-specific guidelines must be consulted (if they exist) to establish a cut-off that takes disease prevalence into account.
<hr>Equivalent germline code (Richards et al., 2015)[7]: BA1/BS1

Evidence criteria combination guidance:
If B1 is applied (at any level) then O4 should <u>not</u> also be used.</td></tr>
<tr><td rowspan="2">B2</td><td>Variant does not fit the mode of action of the gene for the cancer under investigation</td><td>Stand-alone VUS</td></tr>
<tr><td colspan="2">Guidance:

Mode of action should be determined from the Cancer Gene Census[24]. Analysis of the mutational landscape of the gene in the cancer type under investigation through analysis of large datasets (e.g. GENIE data displayed in the annotation track on cBioPortal) or literature searches can also be useful when establishing the method through which oncogenic variants exert their effect.

This code may be applied for null variants in a gene known to act as an oncogene in the cancer type under investigation.

This code may be applied for activating variants in a gene known to act as a tumour suppressor gene in the cancer type under investigation.

Where mode of action of the gene is ambiguous then it is critical the mode of action for the cancer type under investigation is determined.

The presence of a high number of null variants further upstream in the gnomAD database may be an indication that this is not the mechanism of action of the gene. However, it is important to bear in mind that a high incidence of null variants in gnomAD may also be due to CHIP rather than an indicator of disease mechanism.

Please note that where B1 has been used as stand-alone benign evidence (gnomAD frequency >1%), B2 does NOT override this classification.
<hr>Equivalent germline code (Richards et al., 2015)[7]: None (although similar to BP1)</td></tr>
</table>

B3	**Computational evidence DOES NOT support a deleterious effect on the gene or gene product (conservation, evolutionary, splicing impact).**	Supporting [-1]
	Guidance: <u>Missense variants:</u> ● For predicting the impact of missense variants, it is recommended that a meta-predictor tool, such as REVEL is used[14]. ● A REVEL score of ≤0.4 provides evidence against oncogenicity and B3 can be applied at a SUPPORTING level[†]. ● If gene-specific guidance is available, then this should be followed, and evidence can be applied at the strength as specified by the appropriate guidelines. ● The accuracy of REVEL for non-loss of function (non-LOF) variants has not been fully validated; however, Gerasimavicius *et al*., (2022)[25] and Hopkins *et al*., (2023)[26] found that that nearly all computational variant effect predictors underperform on non-LOF mutations, even those based solely on sequence conservation. **It is therefore recommended that REVEL should NOT be used as evidence of benignity for non-LOF (e.g. activating) variants.** [†]Threshold based on analysis of a cohort of 321 pathogenic (ACMG 4/5) and 1238 benign (ACMG 1/2) variants classified by the Leeds Genetics Laboratory to achieve a sensitivity of ~95%. This threshold is also supported by the study of Cubuk *et al*. (2021)[16]. <u>Splicing variants:</u> ● *In silico* splicing prediction tools can be used as evidence to support a significant impact on splicing potential for splice site variants outside the canonical splice acceptor (-1 and -2) and donor (+1 and +2) regions. ● The use of SpliceAI is recommended and B3 may be used at a SUPPORTING level for variants where **all** donor/acceptor loss Δ score(s) ≤0.1[3]. Where gene-specific ClinGen VCEP guidance is available, thresholds from these guidelines can be used; however, B3 can only be applied at a maximum strength of SUPPORTING regardless of gene-specific guidance permitting the use of the equivalent code at higher levels of strength. **Equivalent germline code (Richards *et al.*, 2015)[7]:** BP4 **Evidence criteria combination guidance is detailed in Supplementary Figure 2. Non-permitted combinations exist for this criterion.**	
B4 / **B4 (RNA)**	**Synonymous/intronic variants (with no impact on splicing determined by *in silico* prediction or through *in vitro* splicing assays)**	Strong [-4] Supporting [-1]
	Guidance: <u>Use of *in silico* data to predict impact on splicing (B4_supporting)</u> B4 can be applied at **supporting** for synonymous variants, intronic variants (at or beyond +7/-21) and non-coding variants in the untranslated regions (UTRs) where *in silico* tools predict no impact on the splice consensus sequence nor the creation of a new splice site (i.e. B3 is applicable). Please note this code should be used with caution for synonymous variants in proximity to splice sites, specifically those located at the first base or last three bases of the exon, as there is a higher likelihood	

	of impact on splicing due to disruption of the acceptor or donor motifs[3] (e.g. *TP53* variants affecting the last base of exon 4, exon 6 and exon 9). Synonymous variants with no predicted effect on splicing can be classified as likely benign (B3 and B4). <u>Use of *in vitro* splicing assays to assess impact on splicing [B4 (RNA) strong]</u> Following the recommendations of the ClinGen SVI Splicing Subgroup[3] for re-purposing of the equivalent germline code (BP7), **B4 (RNA)** can be used when a synonymous (silent) or intronic substitution (including intronic variants within the splice donor/acceptor region) is confirmed to have no impact on splicing by a suitable *in vitro* assay, B4 can be upweighted to B4 _Strong (RNA)[3]. The experimental data must be assessed to determine whether the assay performed was of sufficient quality to be used as evidence supportive of the application of the B4 (RNA) code. Considerations include assay design, RNA source, methodology, technology, quantification measurement and use of controls; a more comprehensive list of considerations is included in Table S9, Walker *et al.,* (2023)[3]. Additional validation requirements are available in the CanVIG-UK Consensus Specification (v3.0) [6]. **Equivalent germline code (Richards *et al.*, 2015)[7]:** BP7 **Evidence criteria combination guidance is detailed in Supplementary Figure 2. Non-permitted combinations and notes of caution exist for this criterion**	
B5	**In-frame deletion/insertion in a repetitive region with unknown function**	Supporting [-1]
	Guidance: Application suitable for in-frame deletions/insertions in repetitive regions or regions that are not well conserved in evolution. Consider use of O9 if in-frame deletion/insertion is not present in a repeat region. **Equivalent germline code (Richards *et al.*, 2015)[7]:** BP3	
B6	**Well-established *in vitro* or *in vivo* functional studies demonstrating no damaging effect on protein function**	Strong [-4] Moderate [-2] Supporting [-1]
	Guidance: This code may be applied where *in vivo* and *in vitro* functional studies show no effect of the specific variant in the cancer type under investigation. **Applicable for protein assays only**; for RNA splicing assays see B4 (RNA). **This code is applied at variant level only**. Where functional data, for example from biochemical testing, provides support at the gene rather than variant level, this cannot be used in the variant biological annotation (but may be incorporated when assessing clinical actionability). **In the absence of gene-specific guidance, apply B6 according to CanVIG-UK Consensus Specification[6], where the germline evidence code BS3 should be considered analogous to the SVIG-**	

	UK evidence code B6. In accordance with this, clinical validity of functional evidence and consequential weighting of B6 should be determined according to ClinGen SVI recommendations[22] . **In the case of availability of data from multiple assays:** use guidelines and combination tables described by the CanVIG-UK Consensus Specification[6] when considering assays of variable evidence strength and/or combining concordant/discordant functional assay results. Of note, conflicting functional evidence should not be used in the interpretation of the variant. B6 should only be applied where functional studies have been performed. *In silico* evidence should be assessed using B3. **Equivalent germline code (Richards *et al.*, 2015)[7]:** BS3 **Evidence criteria combination guidance is detailed in Supplementary Figure 2. Non-permitted combinations, restriction and a note of caution exist for this criterion**	
B7	**Tumour phenotype**	Moderate [-2] Supporting [-1]
	Application should be in accordance with gene-specific ClinGen VCEP guidelines (where available) but limited to the level of strength as permitted in these guidelines. This criterion enables other findings, including those from non-genetic disciplines, to be used as evidence in the classification of somatic variants. Specific guidance for the use of mismatch repair (MMR), loss of heterozygosity (LOH), genomic instability scores for classification of *BRCA1/2* variants and tumour mutational burden (TMB) for classification of *POLE* variants is provided in Supplementary Table 4. **Equivalent germline code (Richards *et al.*, 2015)[7]:** None	

References

1. Abou Tayoun AN, Pesaran T, DiStefano MT, *et al*. Recommendations for interpreting the loss of function PVS1 ACMG/AMP variant criterion. *Hum Mutat*. 2018;**39**:1517-1524. doi:10.1002/humu.23626

2. Horak P, Griffith M, Danos AM, *et al*. Standards for the classification of pathogenicity of somatic variants in cancer (oncogenicity): Joint recommendations of Clinical Genome Resource (ClinGen), Cancer Genomics Consortium (CGC), and Variant Interpretation for Cancer Consortium (VICC). *Genetics in Medicine*. 2022;**24**:986-998. doi:10.1016/j.gim.2022.01.001

3. Walker LC, Hoya M de la, Wiggins GAR, *et al*. Using the ACMG/AMP framework to capture evidence related to predicted and observed impact on splicing: Recommendations from the ClinGen SVI Splicing Subgroup. *Am J Hum Genet*. 2023;**110**:1046-1067. doi:10.1016/j.ajhg.2023.06.002

4. Lindeboom RGH, Supek F, Lehner B. The rules and impact of nonsense-mediated mRNA decay in human cancers. *Nat Genet*. 2016;**48**:1112-1118. doi:10.1038/ng.3664

5. Lin JH, Masson E, Boulling A, *et al*. 5′ splice site GC>GT and GT>GC variants differ markedly in terms of their functionality and pathogenicity. *Hum Mutat*. 2020;**41**:1358-1364. doi:https://doi.org/10.1002/humu.24029

6. Garrett A, Allen S, Loong L, *et al*. CanVIG-UK Consensus Specification for Cancer Susceptibility Genes of ACGS Best Practice Guidelines for Variant Classification (v3.0). 2024. Available: https://www.cangene-canvaruk.org/_files/ugd/ed948a_ca951fe798294f7f8545d99ec0909e35.pdf

7. Richards S, Aziz N, Bale S, *et al*. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genetics in Medicine*. 2015;**17**:405-424. doi:10.1038/gim.2015.30

8. Kar SP, Quiros PM, Gu M, *et al*. Genome-wide analyses of 200,453 individuals yield new insights into the causes and consequences of clonal hematopoiesis. *Nat Genet*. 2022;**54**:1155-1166. doi:10.1038/s41588-022-01121-z

9. van Zeventer IA, de Graaf AO, Salzbrunn JB, *et al*. Evolutionary landscape of clonal hematopoiesis in 3,359 individuals from the general population. *Cancer Cell*. 2023;**41**:1017-1031.e4. doi:10.1016/j.ccell.2023.04.006

10. Pejaver V, Byrne AB, Feng BJ, *et al*. Calibration of computational tools for missense variant pathogenicity classification and ClinGen recommendations for PP3/BP4 criteria. *Am J Hum Genet*. 2022;**109**:2163-2177. doi:10.1016/j.ajhg.2022.10.013

11. Safaei S, Baradaran B, Hagh MF, *et al*. Double sword role of EZH2 in leukemia. *Biomedicine & Pharmacotherapy*. 2018;**98**:626-635. doi:https://doi.org/10.1016/j.biopha.2017.12.059

12. Eich ML, Athar M, Ferguson III JE, Varambally S. EZH2-Targeted Therapies in Cancer: Hype or a Reality. *Cancer Res*. 2020;**80**:5449-5458. doi:10.1158/0008-5472.CAN-20-2147

13. Kim KH, Roberts CWM. Targeting EZH2 in cancer. *Nat Med*. 2016;**22**:128-134. doi:10.1038/nm.4036

14. Ioannidis NM, Rothstein JH, Pejaver V, *et al*. REVEL: An Ensemble Method for Predicting the Pathogenicity of Rare Missense Variants. *Am J Hum Genet*. 2016;**99**:877-885. doi:10.1016/j.ajhg.2016.08.016

15. Gunning AC, Fryer V, Fasham J, *et al*. Assessing performance of pathogenicity predictors using clinically relevant variant datasets. *J Med Genet*. 2021;**58**:547. doi:10.1136/jmedgenet-2020-107003

16. Cubuk C, Garrett A, Choi S, *et al*. Clinical likelihood ratios and balanced accuracy for 44 in silico tools against multiple large-scale functional assays of cancer susceptibility genes. *Genetics in Medicine*. 2021;**23**:2096-2104. doi:10.1038/s41436-021-01265-z

17. Wai HA, Lord J, Lyon M, *et al*. Blood RNA analysis can increase clinical diagnostic rate and resolve variants of uncertain significance. *Genetics in Medicine*. 2020;**22**:1005-1014. doi:10.1038/s41436-020-0766-9

18. Rowlands C, Thomas HB, Lord J, *et al*. Comparison of in silico strategies to prioritize rare genomic variants impacting RNA splicing for the diagnosis of genomic disorders. *Sci Rep*. 2021;**11**:20607. doi:10.1038/s41598-021-99747-2

19. Walsh MF, Ritter DI, Kesserwan C, *et al*. Integrating somatic variant data and biomarkers for germline variant classification in cancer predisposition genes. *Hum Mutat*. 2018;**39**:1542-1552. doi:https://doi.org/10.1002/humu.23640

20. Chang MT, Asthana S, Gao SP, *et al*. Identifying recurrent mutations in cancer reveals widespread lineage diversity and mutational specificity. *Nat Biotechnol*. 2016;**34**:155-163. doi:10.1038/nbt.3391

21. Chang MT, Bhattarai TS, Schram AM, *et al*. Accelerating Discovery of Functional Mutant Alleles in Cancer. *Cancer Discov*. 2018;**8**:174-183. doi:10.1158/2159-8290.CD-17-0321

22. Brnich SE, Abou Tayoun AN, Couch FJ, *et al*. Recommendations for application of the functional evidence PS3/BS3 criterion using the ACMG/AMP sequence variant interpretation framework. *Genome Med*. 2019;**12**:3. doi:10.1186/s13073-019-0690-2

23. Ghosh R, Harrison SM, Rehm HL, Plon SE, Biesecker LG, Group on behalf of CSVIW. Updated recommendation for the benign stand-alone ACMG/AMP criterion. *Hum Mutat*. 2018;**39**:1525-1530. doi:https://doi.org/10.1002/humu.23642

24. Sondka Z, Bamford S, Cole CG, Ward SA, Dunham I, Forbes SA. The COSMIC Cancer Gene Census: describing genetic dysfunction across all human cancers. *Nat Rev Cancer*. 2018;**18**:696-705. doi:10.1038/s41568-018-0060-1

25. Gerasimavicius L, Livesey BJ, Marsh JA. Loss-of-function, gain-of-function and dominant-negative mutations have profoundly different effects on protein structure. *Nat Commun*. 2022;**13**:3895. doi:10.1038/s41467-022-31686-6

26. Hopkins JJ, Wakeling MN, Johnson MB, Flanagan SE, Laver TW. REVEL Is Better at Predicting Pathogenicity of Loss-of-Function than Gain-of-Function Variants. *Hum Mutat*. 2023:8857940. doi:https://doi.org/10.1155/2023/8857940

Supplementary Table 3 – O1 SVIG-UK Canonical Variants List

Variants met the following criteria: oncogenic classification (reviewed and agreed by at least two SVIG-UK scientists); highly recurrent and enriched in tumour databases (O4_strong). By exception, a limited number or variants that did not achieve an SVIG-UK score of 10 and were therefore classified as likely oncogenic, but which are well-established, clinically significant variants in specific tumour types, are included in the list according to the expert opinion of SVIG-UK scientists. The terminology used to describe these variants (e.g. canonical) is at the discretion of individual laboratories.

Gene	MANE Select Transcript	Variant	SVIG-UK assessment
ALK	NM_004304.5	c.3520T>C/c.3522C>G/A p.(Phe1174Leu) c.3824G>A p.(Arg1275Gln)	Oncogenic
BRAF	NM_004333.6	c.1406G>C p.(Gly469Ala) c.1781A>G p.(Asp594Gly) c.1798G>A p.(Val600Met) c.1798_1799delinsAA p.(Val600Lys) c.1798_1799delinsAG p.(Val600Arg) c.1799T>A/1799_1800delinsAA p.(Val600Glu) c.1801A>G p.(Lys601Glu)	Oncogenic
CALR	NM_004343.4	c.1099_1150del p.(Leu367ThrfsTer46) (Type 1) c.1154_1155insTTGTC p.(Lys385AsnfsTer47) (Type 2)	Likely Oncogenic
CD79B	NM_000626.4	c.586T>A p.(Tyr196Asn) c.586T>C p.(Tyr196His) c.586T>G p.(Tyr196Asp) c.587A>C p.(Tyr196Ser) c.587A>G p.(Tyr196Cys) c.587A>T p.(Tyr196Phe)	Likely Oncogenic
CSF3R	NM_000760.4	c.1853C>T p.(Thr618Ile)	Oncogenic
CXCR4	NM_003467.3	c.1013C>A/G p.(Ser338Ter)	Likely Oncogenic
DDX41	NM_016222.4	c.1574G>A p.(Arg525His)	Oncogenic
DNMT3A	NM_022552.5	c.2644C>T p.(Arg882Cys) c.2645G>A p.(Arg882His)	Oncogenic
EGFR	NM_005228.5	c.2155G>A p.(Gly719Ser) c.2155G>T p.(Gly719Cys) c.2156G>C p.(Gly719Ala) c.2303G>T p.(Ser768Ile) c.2369C>T p.(Thr790Met) c.2573T>G p.(Leu858Arg) c.2582T>A p.(Leu861Gln)	Oncogenic
EZH2	NM_004456.5	c.1936T>A p.(Tyr646Asn) c.1936T>C p.(Tyr646His) c.1937A>C p.(Tyr646Ser) c.1937A>G p.(Tyr646Cys)	Oncogenic

EZH2 cont	NM_004456.5	c.1937A>T p.(Tyr646Phe)	Oncogenic
FLT3	NM_004119.3	Exon 14 region IF-dup (ITD) c.2503G>C p.(Asp835His) c.2503G>T p.(Asp835Tyr) c.2504A>T p.(Asp835Val) c.2505T>A/G p.(Asp835Glu) c.2508_2510del p.(Ile836del)	Oncogenic
GNAS	NM_000516.7	c.601C>T p.(Arg201Cys) c.602G>A p.(Arg201His)	Oncogenic
H3-3A	NM_002107.7	c.83A>T p.(Lys28Met) c.103G>A/C p.(Gly35Arg) c.103G>T p.(Gly35Trp)	Oncogenic
H3-3B	NM_005324.5	c.110A>T/c.111G>C p.(Lys37Met)	Oncogenic
HRAS	NM_005343.4	c.34G>A p.(Gly12Ser) c.34G>T p.(Gly12Cys) c.35G>A p.(Gly12Asp) c.35G>T p.(Gly12Val) c.37G>A p.(Gly13Ser) c.37G>C p.(Gly13Arg) c.37G>T p.(Gly13Cys) c.38G>A p.(Gly13Asp) c.38G>T p.(Gly13Val) c.181C>A p.(Gln61Lys) c.182A>G p.(Gln61Arg) c.182A>T p.(Gln61Leu) c.183G>C/T p.(Gln61His)	Oncogenic
IDH1	NM_005896.4	c.394C>A p.(Arg132Ser) c.394C>G p.(Arg132Gly) c.394C>T p.(Arg132Cys) c.395G>A p.(Arg132His) c.395G>T p.(Arg132Leu)	Oncogenic
IDH2	NM_002168.4	c.418C>T p.(Arg140Trp) c.419G>A p.(Arg140Gln) c.419G>T p.(Arg140Leu) c.515G>A p.(Arg172Lys) c.515G>T p.(Arg172Met) c.516G>C/T p.(Arg172Ser)	Oncogenic
JAK2	NM_004972.4	c.1849G>T p.(Val617Phe)	Oncogenic
KIT	NM_000222.3	c.1727T>C p.(Leu576Pro) c.2446G>C p.(Asp816His) c.2446G>T p.(Asp816Tyr) c.2447A>T p.(Asp816Val)	Oncogenic
		c.2466T>A/G p.(Asn822Lys)	Likely Oncogenic

KRAS	NM_004985.5	c.34G>A p.(Gly12Ser) c.34G>C p.(Gly12Arg) c.34G>T p.(Gly12Cys) c.34_35delinsTT p.(Gly12Phe) c.35G>A p.(Gly12Asp) c.35G>C p.(Gly12Ala) c.35G>T p.(Gly12Val) c.37G>A p.(Gly13Ser) c.37G>C p.(Gly13Arg) c.37G>T p.(Gly13Cys) c.38G>A p.(Gly13Asp) c.38G>T p.(Gly13Val) c.181C>A p.(Gln61Lys) c.182A>T p.(Gln61Leu) c.182A>G p.(Gln61Arg) c.183A>C/T p.(Gln61His) c.436G>A p.(Ala146Thr)	Oncogenic
MPL	NM_005373.3	c.1514G>A p.(Ser505Asn) c.1543_1544delinsAA p.(Trp515Lys) c.1543_1545delinsAAA p.(Trp515Lys) c.1544G>T p.(Trp515Leu)	Oncogenic
MYD88	NM_002468.5	c.755T>C p.(Leu252Pro)	Oncogenic
NPM1	NM_002520.7	c.860_863dup p.(Trp288CysfsTer12) c.863_864ins(4 bases) p.(Trp288CysfsTer12)	Oncogenic
NRAS	NM_002524.5	c.34G>A p.(Gly12Ser) c.34G>C p.(Gly12Arg) c.34G>T p.(Gly12Cys) c.35G>A p.(Gly12Asp) c.35G>C p.(Gly12Ala) c.35G>T p.(Gly12Val) c.37G>A p.(Gly13Ser) c.37G>C p.(Gly13Arg) c.37G>T p.(Gly13Cys) c.38G>A p.(Gly13Asp) c.38G>T p.(Gly13Val) c.181C>A p.(Gln61Lys) c.182A>G p.(Gln61Arg) c.182A>T p.(Gln61Leu) c.183A>C/T p.(Gln61His) c.436G>A p.(Ala146Thr)	Oncogenic
PAX5	NM_016734.3	c.239C>G p.(Pro80Arg)	Oncogenic
PIK3CA	NM_006218.4	c.1035T>A p.(Asn345Lys) c.1624G>A p.(Glu542Lys)	Oncogenic

PIK3CA cont	NM_006218.4	c.1633G>A p.(Glu545Lys) c.3140A>T p.(His1047Leu) c.3140A>G p.(His1047Arg)	Oncogenic
PDGFRA	NM_006206.6	c.2525A>T p.(Asp842Val)	Likely Oncogenic
POLE	NM_006231.4	c.857C>G p.(Pro286Arg) c.1231G>C/T p.(Val411Leu)	Oncogenic
RET	NM_020975.6	c.2753T>C p.(Met918Thr)	Likely Oncogenic
RHOA	NM_001664.4	c.50G>T p.(Gly17Val)	Oncogenic
RUNX1	NM_001754.5	c.601C>T p.(Arg201Ter) c.601G>A p.(Arg201Gln)	Oncogenic
SETBP1	NM_015559.3	c.2602G>A p.(Asp868Asn) c.2608G>A p.(Gly870Ser)	Oncogenic
SF3B1	NM_012433.4	c.1866G>C/T p.(Glu622Asp) c.1873C>T p.(Arg625Cys) c.1874G>A p.(Arg625His) c.1986C>A/G p.(His662Gln) c.1997A>C p.(Lys666Thr) c.1997A>G p.(Lys666Arg) c.1998G>C/T p.(Lys666Asn) c.2098A>G p.(Lys700Glu) c.2225G>A p.(Gly742Asp)	Oncogenic
SRSF2	NM_001195427.2	c.284C>A p.(Pro95His) c.284C>G p.(Pro95Arg) c.284C>T p.(Pro95Leu) c.284_307del p.(Pro95_Arg102del)	Oncogenic
STAT3	NM_139276.3	c.1919A>T p.(Tyr640Phe) c.1981G>T p.(Asp661Tyr) c.1982A>T p.(Asp661Val)	Oncogenic
TERT	NM_198253.3	c.-124C>T c.-146C>T	Oncogenic
TP53	NM_000546.6	c.524G>A p.(Arg175His) c.659A>G p.(Tyr220Cys) c.733G>A p.(Gly245Ser) c.742C>T p.(Arg248Trp) c.743G>T p.(Arg248Gln) c.817C>T p.(Arg273Cys) c.818G>A p.(Arg273His) c.844C>T p.(Arg282Trp)	Oncogenic
U2AF1	NM_006758.3	c.101C>A p.(Ser34Tyr) c.101C>T p.(Ser34Phe) c.470A>C p.(Gln157Pro) c.470A>G p.(Gln157Arg)	Oncogenic
XP01	NM_003400.4	c.1711G>A p.(Glu571Lys)	Oncogenic

Supplementary Table 4– O11/B7 Examples

Please note that these are guiding examples and not a comprehensive list.

<table>
<tr><th colspan="2">Tumour Phenotype</th><th>Criteria/Strength†</th></tr>
<tr><td rowspan="10">Mismatch repair (MMR)</td><td colspan="2">Germline status unknown</td></tr>
<tr><td>• Microsatellite instability high (MSI-H) - DO NOT USE for any gene
• MLH1 variant: Loss of MLH1/PMS2 (MLH1 methylation status unknown) – DO NOT USE</td><td>N/A</td></tr>
<tr><td>• MLH1 variant: Loss of MLH1/PMS2 AND MLH1 methylation normal
• MSH2 or MSH6 variant: Loss of MSH2/MSH6 or MSH6 only loss (MSH6 variant)
• PMS2 variant – Loss of MLH1/PMS2 or PMS2 only
• Informative LOH at the relevant locus for the variant</td><td>O11 Supporting [+1]</td></tr>
<tr><td colspan="2">Germline status known (no MMR variants identified)</td></tr>
<tr><td>• MSI-H
• Informative LOH at the relevant locus for the variant</td><td>O11 Supporting [+1]</td></tr>
<tr><td>• MLH1 variant: Loss of MLH1/PMS2 AND MLH1 methylation normal
• MSH2 or MSH6 variant: Loss of MSH2/MSH6 or MSH6 only loss (MSH6 variant)
• PMS2 variant – Loss of MLH1/PMS2 or PMS2 only</td><td>O11 Moderate [+2]</td></tr>
<tr><td colspan="2">Germline status known (relevant 1st hit identified)</td></tr>
<tr><td>• Do not use any evidence points for MSI/immunohistochemistry (IHC)</td><td>N/A</td></tr>
<tr><td colspan="2">Notes:
• MSI/IHC – evidence cannot be combined
• LOH can be combined with MSI/IHC</td></tr>
<tr></tr>
<tr><td rowspan="2">Loss-of-heterozygosity (LOH) for any TSG</td><td>• Germline status unknown</td><td>O11 Supporting [+1]</td></tr>
<tr><td>• No germline variants identified</td><td>O11 Moderate [+2]</td></tr>
<tr><td>Genes where biallelic inactivation due to concurrent germline and somatic variants is a well-established mechanism (e.g. DDX41), and</td><td>Pathogenic/likely pathogenic, confirmed/putative germline variant has been detected in a gene where concurrent germline and somatic variants are a well-established, frequent mechanism of biallelic inactivation (e.g. DDX41[1,2]). Points can be applied to the second</td><td>O11 moderate [+2]</td></tr>
</table>

P/LP variant highly likely to be germline has been identified	(somatic) hit.	
	Pathogenic/likely pathogenic confirmed/putative germline variant has been detected in a gene where concurrent germline and somatic variants are a recognised, but infrequent, mechanism of biallelic inactivation (e.g. *CEBPA*). Point can be applied to the second (somatic) hit.	O11 Supporting [+1]
	Notes: SVIG-UK advise this evidence is applied with caution. At the time of classifying variants identified in tumour, germline material is usually unavailable. However, the presence of a variant with an allele frequency consistent with germline status, that would be classified as pathogenic or likely pathogenic if confirmed to be germline, may provide evidence in classification of a potential somatic second hit. Germline classifications are well-established and stringent justifying their use in the evidence criteria.	
Tumour mutational burden (TMB) for *POLE*	• TMB high	O11 Supporting [+1]
	• TMB low	B6 Moderate [+2]
	Notes: Variants in *POLE* must be located within the exonuclease domain to apply the above criteria[3].	

References

1. Kim K, Ong F, Sasaki K. Current Understanding of DDX41 Mutations in Myeloid Neoplasms. *Cancers (Basel)*. 2023;**15**:344. doi:10.1038/nm.4036

2. Maierhofer A, Mehta N, Chisholm RA, *et al*. The clinical and genomic landscape of patients with DDX41 variants identified during diagnostic sequencing. *Blood Adv*. 2023;**7**:7346-7357. doi:10.1182/bloodadvances.2023011389

3. Dong S, Zakaria H, Hsiehchen D. Non-Exonuclease Domain POLE Mutations Associated with Immunotherapy Benefit. *Oncologist*. 2022;**27**:159-162. doi:10.1093/oncolo/oyac017

Supplementary Table 5 – Key Literature Databases and Tools

Evidence criteria		References/Reference Dataset/In Silico Tool Listed	Website Link
O1	**Canonical Variant**	N/A	
O2	**Null variant in a tumour suppressor gene (TSG)**	Jaganathan et al., (2019). PMID: 30661751	Predicting Splicing from Primary Sequence with Deep Learning - PubMed
		SpliceAI	SpliceAI Lookup
		Tayoun et al., (2018). PMID: 30192042	Recommendations for interpreting the loss of function PVS1 ACMG/AMP variant criterion - PubMed
		Horak et al., (2022). PMID: 35101336.	Standards for the classification of pathogenicity of somatic variants in cancer (oncogenicity): Joint recommendations of Clinical Genome Resource (ClinGen), Cancer Genomics Consortium (CGC), and Variant Interpretation for Cancer Consortium (VICC) - PubMed
		Walker et al., 2023 PMID: 37352859	Using the ACMG/AMP framework to capture evidence related to predicted and observed impact on splicing: Recommendations from the ClinGen SVI Splicing Subgroup - PubMed
		Garrett et al., (2024). CanVIG-UK Consensus Specification for Cancer Susceptibility Genes (v3.00)	CanVIG-UK Consensus Specifications | CanGene-CanVar
		MMR (CanVar/CanVIG)	CanVIG Gene Guidance | CanGene-CanVar
		RUNX1 (ClinGen VCEP)	ClinGen Myeloid Malignancy Variant Curation Expert Panel Specifications
		BRCA1/BRCA2 (CanVar/CanVIG)	CanVIG Gene Guidance | CanGene-CanVar
		Lin et al., (2020). PMID: 32369867	5' splice site GC>GT and GT>GC variants differ markedly in terms of their functionality and pathogenicity
		Lindeboom et al., (2016). PMID: 27618451	The rules and impact of nonsense-mediated mRNA decay in human cancers - PubMed
O3	**Absent or very rare in population database**	Genome Aggregation Database (gnomAD) v4.1	gnomAD
		Kar SP et al., (2022). PMID: 35835912	Genome-wide analyses of 200,453 individuals yield new insights into the causes and consequences of clonal hematopoiesis - PubMed
		Van Zeventer et al., (2023). PMID: 37146604	Evolutionary landscape of clonal hematopoiesis in 3,359 individuals from the general population
		Pejaver et al. (2022). PMID: 36413997	Calibration of computational tools for missense variant pathogenicity classification and ClinGen recommendations for PP3/BP4 criteria - PubMed

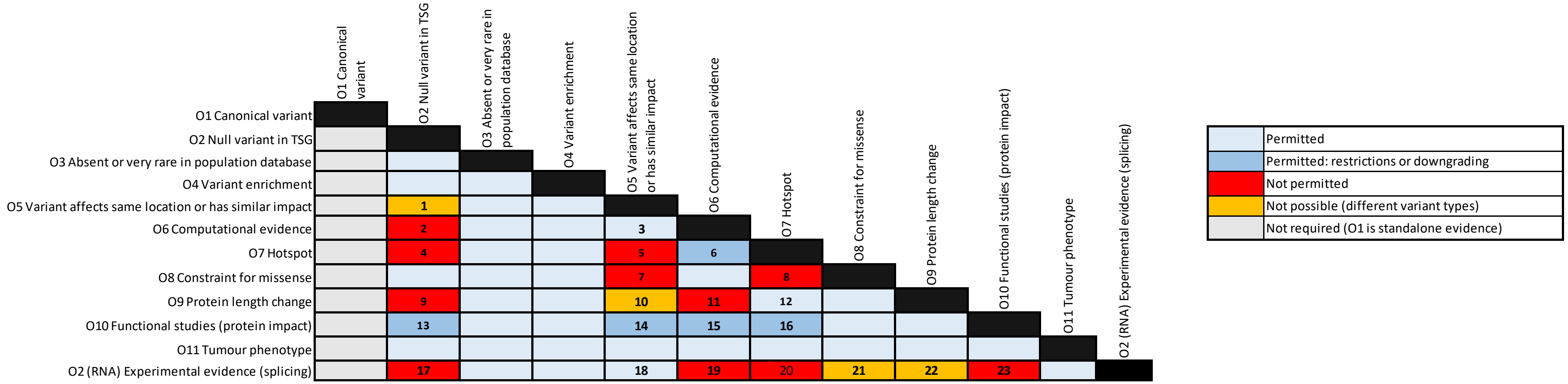
		CanVIG-UK	CanVIG-UK Consensus Specifications | CanGene-CanVar
		ACGS 2024	uk-practice-guidelines-for-variant-classification-v12-2024.pdf
		ACMG/AMP ClinGen variant curation expert panel (VCEP)	ClinGen Variant Curation Expert Panel (VCEP) - Criteria Specification Registry
		ClinGen Sequence Variant Interpretation (SVI) Working Group	Sequence Variant Interpretation - ClinGen | Clinical Genome Resource
		Gene-specific guidelines	CanVIG Gene Guidance | CanGene-CanVar
O4	**Enriched in a somatic variant database**	COSMIC	COSMIC | Catalogue of Somatic Mutations in Cancer
		GENIE	cBioPortal (requires sign-in)
		Froyen et al., (2019). PMID: 31888289	Standardization of Somatic Variant Classifications in Solid and Haematological Tumours by a Two-Level Approach of Biological and Clinical Classes: An Initiative of the Belgian ComPerMed Expert Panel
O5	**Missense at an amino acid where a different missense variant has previously been determined to be oncogenic.**	Ioannidis et al., (2016). PMID: 27666373	REVEL: An Ensemble Method for Predicting the Pathogenicity of Rare Missense Variants - PubMed
		REVEL	REVEL score download
		Pejaver et al. (2022). PMID: 36413997	Calibration of computational tools for missense variant pathogenicity classification and ClinGen recommendations for PP3/BP4 criteria - PubMed
		Jaganathan et al., (2019). PMID: 30661751	Predicting Splicing from Primary Sequence with Deep Learning - PubMed
		Walker et al., 2023 PMID: 37352859	Using the ACMG/AMP framework to capture evidence related to predicted and observed impact on splicing: Recommendations from the ClinGen SVI Splicing Subgroup - PubMed
		SpliceAI	SpliceAI Lookup
O6	**Computational evidence supports a deleterious effect on the gene or gene product (conservation, evolutionary, splicing impact etc).**	REVEL	REVEL score download
		Ioannidis et al., (2016). PMID: 27666373	REVEL: An Ensemble Method for Predicting the Pathogenicity of Rare Missense Variants - PubMed
		Gunning et al., (2020). PMID: 32843488	Assessing performance of pathogenicity predictors using clinically relevant variant datasets - PubMed
		de Sainte Agathe et al., (2023). PMID: 36765386	SpliceAI-visual: a free online tool to improve SpliceAI splicing variant interpretation - PMC
		Cubuk et al., (2021). PMID: 34230640	Clinical likelihood ratios and balanced accuracy for 44 in silico tools against multiple large-scale functional assays of cancer susceptibility genes - PubMed
		SpliceAI	SpliceAI Lookup

		Pejaver et al. (2022). PMID: 36413997	Calibration of computational tools for missense variant pathogenicity classification and ClinGen recommendations for PP3/BP4 criteria - PubMed
		Wai et al., (2020). PMID: 32123317	Blood RNA analysis can increase clinical diagnostic rate and resolve variants of uncertain significance - PubMed
		Rowlands et al., (2021). PMID: 34663891	Comparison of in silico strategies to prioritize rare genomic variants impacting RNA splicing for the diagnosis of genomic disorders - PubMed
		Walker et al., 2023 PMID: 37352859	Using the ACMG/AMP framework to capture evidence related to predicted and observed impact on splicing: Recommendations from the ClinGen SVI Splicing Subgroup - PubMed
		ACMG/AMP ClinGen variant curation expert panel (VCEP)	ClinGen Variant Curation Expert Panel (VCEP) - Criteria Specification Registry
O7	**Located in a mutational hotspot and/or critical and well-established functional domain (e.g. active site of an enzyme) without benign variation**	Cancer Hotspots	Cancer Hotspots
		Decipher	Decipher
		GENIE	cBioPortal (requires sign-in)
		COSMIC	COSMIC | Catalogue of Somatic Mutations in Cancer
		ClinVar Database	ClinVar
		ACMG/AMP ClinGen variant curation expert panel (VCEP)	ClinGen Variant Curation Expert Panel (VCEP) - Criteria Specification Registry
		Chang et al., (2016). PMID: 26619011	Identifying recurrent mutations in cancer reveals widespread lineage diversity and mutational specificity - PubMed
		Chang et al., (2018). PMID: 29247016	Accelerating Discovery of Functional Mutant Alleles in Cancer - PubMed
		Genome Aggregation Database (gnomAD) v4.1	gnomAD
O8	**Overall constraint for missense variation, at the level of the gene or domain/region, where missense variants are a common mechanism of disease.**	Genome Aggregation Database (gnomAD) v2.2.1	
		MetaDome	MetaDome web server
		Decipher	Decipher
O9	**Protein Length change due to in-frame deletions/insertions in a non-repeat region or stop-loss variants**	ACMG/AMP ClinGen variant curation expert panel (VCEP)	ClinGen Variant Curation Expert Panel (VCEP) - Criteria Specification Registry

O10	**Well-established in vitro or in vivo functional studies demonstrating a functionally abnormal result consistent with the mechanism of disease**	Brnich et al., (2020). PMID: 31892348	Recommendations for application of the functional evidence PS3/BS3 criterion using the ACMG/AMP sequence variant interpretation framework - PubMed
		Garrett et al., (2024). CanVIG-UK Consensus Specification for Cancer Susceptibility Genes (v3.0)	CanVIG-UK Consensus Specifications | CanGene-CanVar
O11	**Tumour phenotype**	ACMG/AMP ClinGen variant curation expert panel (VCEP)	ClinGen Variant Curation Expert Panel (VCEP) - Criteria Specification Registry
		Kim et al., (2023). PMID: 36672294	Current Understanding of DDX41 Mutations in Myeloid Neoplasms - PubMed
		Maierhofer et al, (2023). PMID: 37874914	The clinical and genomic landscape of patients with DDX41 variants identified during diagnostic sequencing - PubMed
B1	**Variant is present at a high level in a general population database**	Genome Aggregation Database (gnomAD) v4.1	gnomAD
		ACMG/AMP ClinGen variant curation expert panel (VCEP)	ClinGen Variant Curation Expert Panel (VCEP) - Criteria Specification Registry
B2	**Variant does not fit the mode of action of the gene for the cancer under investigation**	COSMIC Cancer Gene Census	Cancer Gene Census
		GENIE	cBioPortal (requires sign-in)
		Sondka et al., (2018). PMID: 30293088	The COSMIC Cancer Gene Census: describing genetic dysfunction across all human cancers - PubMed
B3	**Computational evidence DOES NOT support a deleterious effect on the gene or gene product (conservation, evolutionary, splicing impact).**	Ioannidis et al., (2016). PMID: 27666373	REVEL: An Ensemble Method for Predicting the Pathogenicity of Rare Missense Variants - PubMed
		Hopkins et al., (2023)	REVEL Is Better at Predicting Pathogenicity of Loss-of-Function than Gain-of-Function Variants - Hopkins - 2023 - Human Mutation - Wiley Online Library
		ACMG/AMP ClinGen variant curation expert panel (VCEP)	ClinGen Variant Curation Expert Panel (VCEP) - Criteria Specification Registry
		Gerasimavicius et al., (2022). PMID: 35794153	Loss-of-function, gain-of-function and dominant-negative mutations have profoundly different effects on protein structure - PubMed
		Cubuk et al., (2021). PMID: 34230640	Clinical likelihood ratios and balanced accuracy for 44 in silico tools against multiple large-scale functional assays of cancer susceptibility genes - PubMed
		SpliceAI	SpliceAI Lookup
		Walker et al., 2023 PMID: 37352859	Using the ACMG/AMP framework to capture evidence related to predicted and observed impact on splicing: Recommendations from the ClinGen SVI Splicing Subgroup - PubMed

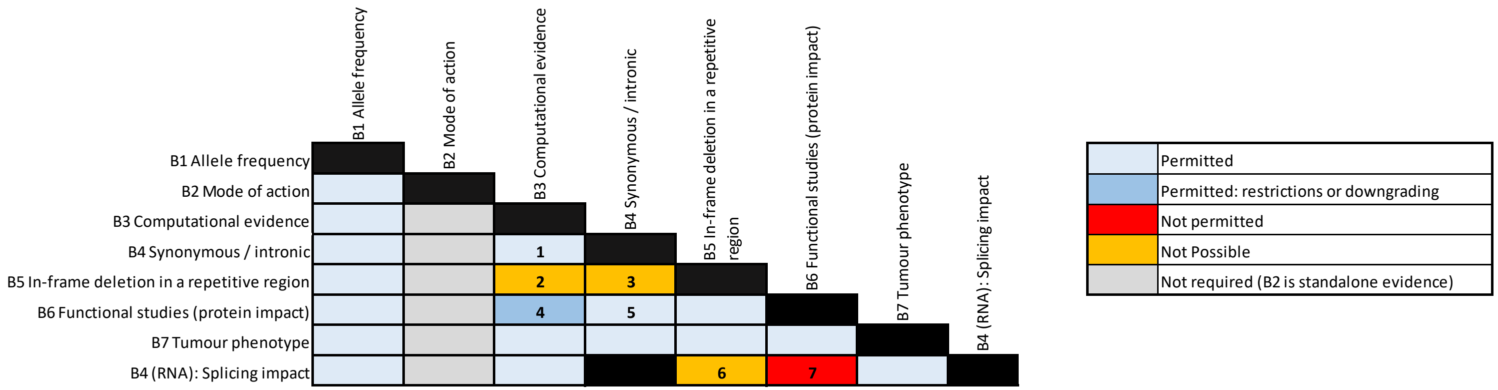
B4	**Synonymous / intronic variants (outside canonical splice sites; at +7/-21 or beyond) for which splicing prediction algorithms predict no impact to the splice consensus sequence nor the creation of a new splice site AND the nucleotide is not highly conserved**	Synonymous Mutations in Cancer Database	SynMICdb
		Walker et al., 2023 PMID: 37352859	Using the ACMG/AMP framework to capture evidence related to predicted and observed impact on splicing: Recommendations from the ClinGen SVI Splicing Subgroup - PubMed
B5	**In-frame deletion in a repetitive region with unknown function**	N/A	
B6	**Well-established in vitro or in vivo functional studies demonstrating no damaging effect on protein function or RNA splicing**	Brnich et al., (2020). PMID: 31892348	Recommendations for application of the functional evidence PS3/BS3 criterion using the ACMG/AMP sequence variant interpretation framework - PubMed
		Garrett et al., (2024). CanVIG-UK Consensus Specification for Cancer Susceptibility Genes (v3.0)	CanVIG-UK Consensus Specifications | CanGene-CanVar
B7	**Tumour phenotype**	ACMG/AMP ClinGen variant curation expert panel (VCEP)	ClinGen Variant Curation Expert Panel (VCEP) - Criteria Specification Registry

Supplementary Figure 1 – Code Combinations Guidance (Oncogenicity)



1	O5	O2	**Not possible:** O5 is reserved for missense variants
2	O6	O2	**Not permitted:** Evidence from additional predictive evidence codes (O6) is not permitted as its effect is assumed in the O2 criterion
3	O6	O5	**Caution:** For application of O5 in the context of splicing; in silico predictive tools (O6) must demonstrate the same predicted impact as the variant under assessment (see O5 section and Walker *et al* 2023 for more information).
4	O7	O2	**Not permitted:** O2 and O7 should not be used together. O7 is for missense/small variants only
5	O7	O5	**Not permitted:** O5 and O7 cannot be applied together; use evidence code which achieves the highest strength. Note, this includes where points are awarded for O5 additional points cannot be awarded for constraint at gene/domain level
6	O7	O6	**Restriction:** Sum of points awarding from combining O6 and O7 is limited to 4 pts (strong) to avoid double counting key attributes of critical domains / residues which are also often measured by *in silico* tools (for example evolutionary conservation)
7	O8	O5	**Not permitted:** Avoid double counting evidence for constraint; O8 should not be used in combination with O5
8	O8	O7	**Not permitted:** Avoid double counting evidence for constraint; O8 should not be used in combination with O7
9	O9	O2	**Not permitted:** O9 is used for in-frame deletions/insertions that are <1 exon in length
10	O9	O5	**Not possible:** O5 is reserved for missense variants
11	O9	O6	**Not permitted:** Prediction of pro-pathogenicity is incorporated within O9 (O6 cannot not be applied additionally)
12	O9	O7	O7 is reserved for missense variants and small insertions/duplications only
13	O10	O2	**Restriction:** O2 and O10 can applied together for missense variants. Note; use of RNA splicing assays are incorporated into O2 (RNA)
14	O10	O5	**Restriction:** O5 and O10 can only be applied together if the reference variant can be classified as (likely) oncogenic/pathogenic without the use of functional studies
15	O10	O6	**Restriction:** Evidence from prediction tools (except for splicing predictions) may be used in combination with evidence from protein impact assays (O10)
16	O10	O7	**Restriction:** O7 can only be used alongside O10 if the hotspot/critical region can be defined as evidence independent of functional studies
17	O2 (RNA)	O2	**Not permitted:** Where O2 (RNA) is applied (for splicing variants), application of additional prediction evidence codes for splicing impact is not permitted
18	O2 (RNA)	O5	**Caution:** For application of O5 in the context of splicing; functional evidence [O2 (RNA)] must demonstrate the same predicted impact as the variant under assessment (see O5 section and Walker *et al* 2023 for more information).
19	O2 (RNA)	O6	**Not permitted:** Where O2 (RNA) is applied (for splicing variants), application of additional prediction evidence codes for splicing impact is not permitted
20	O2 (RNA)	O7	**Not permitted:** Where O2 (RNA) is applied (for splicing variants), application of O7 defining hotspot regions should not be applied
21	O2 (RNA)	O8	**Not possible:** Different variant types
22	O2 (RNA)	O9	**Not possible:** Different variant types
23	O2 (RNA)	O10	**Not permitted:** Use of RNA splicing assays are incorporated into O2 (RNA); O10 cannot be used additionally

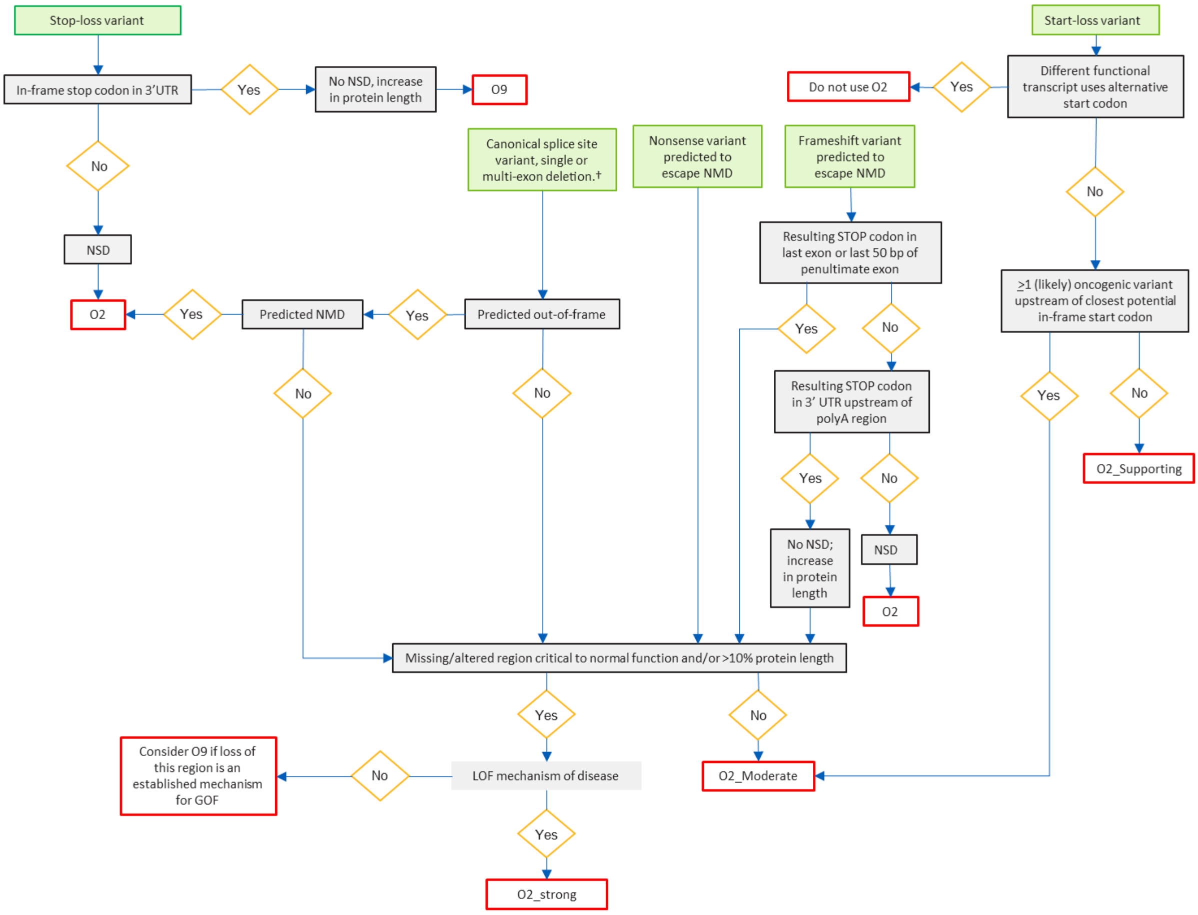
Supplementary Figure 2 – Code Combinations Guidance (Benignity)



1	B4	B3	**Caution:** *In silico* tools must be used to confirm absence of splicing effect (B3) when applying B4.
2	B5	B3	Not possible: B3 cannot be used with B5 (in-frame insertion/deletion within a repetitive or poorly conserved region is recognised through B5).
3	B5	B4	Not possible: B5 cannot be used with B4 as these are different variant types
4	B6	B3	Restriction: B6 should only be used for protein impact. Where *in silico* tools predict an impact on splicing then B6 should not be applied for an assay of protein function.
5	B6	B4	**Caution:** B6 should be used for functional evidence of protein impact only; B4 (RNA) should be used for functional evidence of splicing impact.
6	B4(RNA)	B5	Not possible: B4(RNA) cannot be used with B5 as these are different variant types
7	B4(RNA)	B6	**Not permitted.** B6 is restricted to protein assays. B4 (RNA) should be used for splicing data.

Supplementary Figure 3 - Use of O2/O9 for variants causing protein length changes

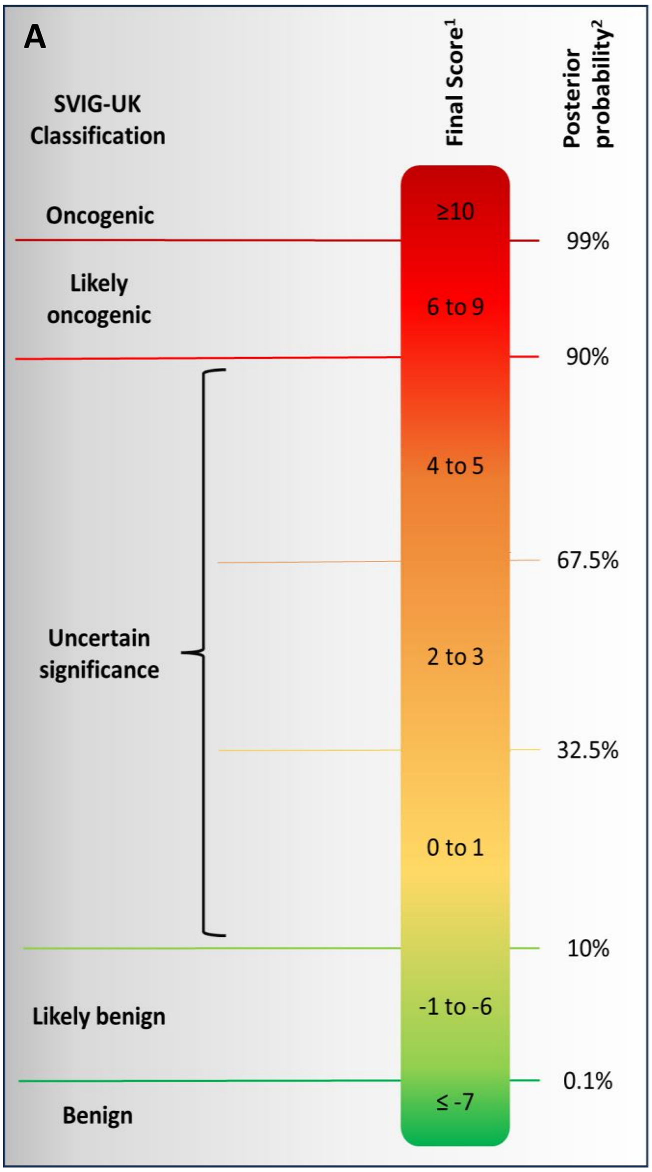
Use of O2 and O9 for start-loss, stop-loss and truncating/frameshift variants in the last exon or last 50 bp of penultimate exon (predicted to escape NMD). Figure adapted from ACGS 2024 PVS1/PM4 decision tree[1]. †Splicing variant should be considered in the context of either: 1) exon skipping, 2) intron retention, 3) use of an alternative splice site. NMD = nonsense mediated decay; NSD = nonstop mediated decay.



References

1. Durkie M, Cassidy EJ, Berry I, *et al*. ACGS Best Practice Guidelines for Variant Classification in Rare Disease 2024 (v1.2). 2024. Available: https://www.acgs.uk.com/media/12533/uk-practice-guidelines-for-variant-classification-v12-2024.pdf

Supplementary Figure 4 –Points-Based System for Oncogenicity Classification. A) Classification point ranges B) Evidence elements weighting C) Classification categories and exponent sums. Figure adapted from UK Best Practice Guidelines for Variant Classification v1.2[1] and Garrett *et al*. (2021)[2].



B

Evidence strength	Oncogenicity exponent score	Benignity exponent score
Very strong	+8	-8
Strong	+4	-4
Moderate	+2	-2
Supporting	+1	-1

C

Classification category	Exponent sum
Oncogenic	≥10
Likely oncogenic	6 to 9
Variant of uncertain significance	0 to 5
Likely Benign	-1 to -6
Benign	≤-7

References

1. Durkie M, Cassidy EJ, Berry I, *et al*. ACGS Best Practice Guidelines for Variant Classification in Rare Disease 2024 (v1.2). 2024. Available: https://www.acgs.uk.com/media/12533/uk-practice-guidelines-for-variant-classification-v12-2024.pdf
2. Garrett A, Durkie M, Callaway A, *et al*. Combining evidence for and against pathogenicity for variants in cancer susceptibility genes: CanVIG-UK consensus recommendations. *J Med Genet* 2021;**58**:297-304. doi: 10.1136/jmedgenet-2020-107248